Number System

SWE 4101
Introduction to Software Engineering

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Outline

- Non-positional number system.
- Positional number system.
- Decimal number system.
- Binary number system.
- Octal number system.
- Hexadecimal number system.

Number System

- **Two types of number system**
  - Non-positional number system.
  - Positional number system.

Non-positional Number System

- **Characteristics**
  - Each symbol represents the same value regardless of its positions in the number.
  - Symbols are just added to determine a particular number.
  - Example: Roman Number System.

I	1
II	2
V	5
VI	6
X	10

  - Difficulty in Arithmetic operations.

Positional Number System

- **Characteristics**
  - Symbols are less, called digits.
  - Digits have different values depending on the position in the number.
  - Example: Decimal Number System.

Positional Number System

- **The value of each digit depends on:**
  - Digit itself (face value).
  - Position in the number.
  - Base of the number system.
- **Value of a number =** $\sum \text{value of } the \text{ digit} * \text{base}^{\text{index-1}}$

  [index is the position of the digit]
- **Base – Total number of different digits in the number system.**
  - For any number system maximum value of the of a single digits always equal to one less than the base.

Decimal Number System

- Positional number system.
- Digits – 0,1,2,3,4,5,6,7,8,9.
- Example:

$$4321_{10} = 1 * 10^0 + 2 * 10^1 + 3 * 10^2 + 4 * 10^3$$
$$= 1 + 20 + 300 + 4000$$

Binary Number System

- Positional number system.
- Digits – 0,1.
- Example:

$$1010_2 = 0 * 2^0 + 1 * 2^1 + 0 * 2^2 + 1 * 2^3$$
$$= 0 + 2 + 0 + 8$$

Octal Number System

- Positional number system.
- Digits – 0,1,2,3,4,5,6,7.
- Example:

$$7654_8 = 4 * 8^0 + 5 * 8^1 + 6 * 8^2 + 7 * 8^3$$
$$= 4 + 40 + 384 + 3584$$
$$= 4012$$

Hexadecimal Number System

- Positional number system.
- Digits – 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F.
- Example:

$$\begin{aligned} ABCD_8 &= 13 * 16^0 + 12 * 16^1 + 11 * 16^2 + 10 * 16^3 \\ &= 13 + 192 + 2816 + 40960 \\ &= 43981 \end{aligned}$$

Converting Any Base To Decimal

- Step 1: Determine the positional value of each digit
- Step 2: Multiply the decimal value of the digit with the positional value.
- Step 3: Summation of these products is the result.

- Value of a number = $\sum decimal\ value\ of\ the\ digit * base^{index-1}$

[index is the position of the digit]

Converting Decimal Value To Any Base

Division-Remainder method

- Step 1: Divide the decimal value with the base.
- Step 2: Record the remainder from the step 1 as the right most digit (least significant digit).
- Step 3: Go to step 1 with the quotient as the new decimal value until the quotient is 0.

Converting Decimal Value To Binary

to Base	Number (Quotient)	Remainder
2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1



$(25)_{10} = (11001)_2$

Fractional Number

- **The positional value of the digit of a number after fractional point is**

$$X = base^{-position}$$

[Position is counted from the fraction point]

Binary Point ↓

Position	4	3	2	1	0	.	-1	-2	-3	-4
Position Value	2^4	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	2^{-4}
Quantity Represented	16	8	4	2	1		$^1/_2$	$^1/_4$	$^1/_8$	$^1/_{16}$

Fractional Number

- Decimal from octal number

$$
\begin{aligned}
127.54_8 &= 1 \times 8^2 + 2 \times 8^1 + 7 \times 8^0 + 5 \times 8^{-1} + 4 \times 8^{-2} \\
&= 64 + 16 + 7 + {}^5/_8 + {}^4/_{64} \\
&= 87 + 0.625 + 0.0625 \\
&= 87.6875_{10}
\end{aligned}
$$

Fractional Number

Decimal to any base

- Step 1: Multiply the fractional number with the Base.
- Step 2: Record the non-fractional part of the resulting number from leftmost digit after fractional point.
- Step 3: Go to step 1 with the fractional part of the resulting number (stop if the fractional part is 0 or if significant number of digits are calculated).

Fractional Number

Convert 0.865 from Base 10 to Base 2,8 and 16.

0.865	0.865	0.865
x 2	x 8	x 16
1.730	6.920	5190
x 2	x 8	865 x
1.460	7.360	13.840
x 2	x 8	x 16
0.920	2.880	5040
x 2	x 8	840 x
1.840	7.040	13.440
x 2		x 16
1.680	The octal equivalent of	2640
x 2	$(0.865)_{10}$ is $(.6727)_8$	440 x
1.360		7.040
The binary equivalent of $(.865)_{10}$ is $(.110111)_2$		The number 13 in hexadecimal is D. The hexadecimal equivalent of $(0.865)_{10}$ is $(.DD7)_{16}$

Conversion among two bases

- Step 1: Convert the number from "From Base" to base 10.
- Step 2: Convert the number in base 10 to "To Base" number.

Shortcut method (Binary to Octal)

$1101010_2 = ?_8$

Step 1: Divide the binary digits into groups of 3 starting from right

001 101 010

Step 2: Convert each group into one octal digit

$001_2 = 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1$

$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5$

$010_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$

Hence, $1101010_2 = 152_8$

Shortcut method (Octal to Binary)

$562_8 = ?_2$

Step 1: Convert each octal digit to 3 binary digits

$5_8 = 101_2$, $6_8 = 110_2$, $2_8 = 010_2$

Step 2: Combine the binary groups

$$562_8 = \underset{5}{\underline{101}} \quad \underset{6}{\underline{110}} \quad \underset{2}{\underline{010}}$$

Hence, $562_8 = 101110010_2$

Shortcut method (Binary to Hexadecimal)

$111101_2 = ?_{16}$

Step 1: Divide the binary digits into groups of four starting from the right

0011 1101

Step 2: Convert each group into a hexadecimal digit

$0011_2 = 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 3_{10} = 3_{16}$

$1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 3_{10} = D_{16}$

Hence, $111101_2 = 3D_{16}$

Shortcut method (Hexadecimal to Binary)

$2AB_{16} = ?_2$

Step 1: Convert each hexadecimal digit to a 4 digit binary number

$2_{16} = 2_{10} = 0010_2$
$A_{16} = 10_{10} = 1010_2$
$B_{16} = 11_{10} = 1011_2$

Step 2: Combine the binary groups

$$2AB_{16} = \underset{2}{\underline{0010}} \quad \underset{A}{\underline{1010}} \quad \underset{B}{\underline{1011}}$$

Hence, $2AB_{16} = 001010101011_2$

Equivalent Chart

Decimal	Binary	Octal	Hexadecimal
0	0000	000	0
1	0001	001	1
2	0010	002	2
3	0011	003	3
4	0100	004	4
5	0101	005	5
6	0110	006	6
7	0111	007	7
8	1000	010	8
9	1001	011	9
10	1010	012	A
11	1011	013	B
12	1100	014	C
13	1101	015	D
14	1110	016	E
15	1111	017	F
16	10000	020	10